# William Liu

647-513-8803 | liu.wllm@gmail.com | linkedin.com/in/liuwllm | github.com/liuwllm | liuwllm.me

## Education

**Western University** **Graduation Date:** June 2027

*Honors in Computer Science and Honors in Business Administration (HBsc. + HBA)* ***GPA**: 3.8/4.0*

- Awards: Dean's Honor List, Western Scholarship of Excellence ($8,000), Andrew & Sarah Hamilton Scholarship ($6,600)
- Relevant Coursework: Data Structures & Algorithms, Software Tools & Systems Programming, Computer Organization & Architecture, Operating Systems, Object-Oriented Design & Analysis

## Work Experience

**Meta** Jun 2026 – Aug 2026

*Software Engineer Intern* *Menlo Park, CA*

- Closed a major visibility gap in failover coverage for Meta's edge infrastructure by designing and deploying a real-time Python service that continuously detects organically rerouted traffic events, surfacing 130+ previously unexamined degradations.
- Automated user impact analyzers for egress and goodput across detected network incident events, cutting assessment time from hours to under 10 minutes by minimizing manual data pulls and ad-hoc investigations performed by engineers.
- Safeguarded user traffic during failover testing by deploying an automated service that ran 3,300+ weekly failover readiness checks across Meta's edge network, confirming fallback locations could absorb rerouted traffic without degrading user media quality.
- Enabled data-driven capacity planning conversations with ISP and network partners (ASNs) by building an internal web dashboard with per-event impact reports quantifying the user impact of analyzed events.

**Major League Hacking** Jun 2025 – Sep 2025

*Site Reliability Engineer (SRE) Fellow* *Remote*

- Reduced the deployment time of a Flask web app by 41% by optimizing Docker containers through layer caching and automating releases with GitHub Actions and Bash, enabling faster shipping with fewer manual errors and less downtime.
- Strengthened system reliability by monitoring Linux hosts and Docker containers with Prometheus and Grafana, building dashboards and alerts that detect container failures and resource spikes within 2 minutes.

**Trojan Technologies** May 2025 – Aug 2025

*Software Engineer Intern* *London, ON*

- Enabled data teams to build analytics and AI models on UV system telemetry by migrating 27% more legacy tables from Azure to Snowflake through automated Python and Airflow pipelines, laying the groundwork for predictive maintenance insights.
- Eliminated gaps in historical device data for an IoT monitoring platform by raising Airflow backfill coverage 53%, restoring data lost to pipeline ingestion failures and improving the accuracy of customer-facing dashboards.
- Streamlined delivery of new ETL pipelines by increasing release frequency from weekly to daily through a Jenkins CI/CD pipeline and Bash automation.

## Leadership

**Lead Web Developer** May 2023 – Jan 2026

*Hack Western* *London, ON*

- Architected the full-stack platform (frontend, backend, database, deployment) for one of Canada's largest collegiate hackathons, powering promotion, applications, and event-day operations for 1,400+ applicants and 300+ accepted participants.
- Led 6 developers through feature planning, code reviews, and task delegation to ship the hackathon's web platform in 8 months.
- Engineered a React-based draggable canvas and interactive elements animated using CSS for hackathon promotional website.

## Projects

**Anishelf** | *TypeScript, Next.js, Flask, PostgreSQL, GraphQL, Docker, nginx*

- Developed a full-stack web application using Next.js and Flask to automatically generate exportable vocabulary flashcard decks for foreign languages based on subtitles extracted from animated series
- Deployed app using Docker containers on a DigitalOcean VPS, utilizing nginx as a reverse proxy to manage traffic

**YACHIP-8 Emulator** | *Rust, SDL2*

- Developed a CHIP-8 emulator using Rust to emulate CHIP-8 system architecture
- Implemented a CPU including the CPU instruction set, registers, and memory management and a front end to display graphics using SDL2

## Technical Skills

**Languages**: Python, TypeScript, JavaScript, Java, Rust, C++, Bash, HTML, CSS
**Frameworks/Libraries**: React, Node.js, Express.js, Next.js, Flask, SQLAlchemy, Selenium, Spring Boot, JUnit
**Tools**: Git, Linux, Unix, Docker, Kubernetes, MongoDB, PostgreSQL, MySQL, GraphQL, AWS, DigitalOcean, nginx, Jenkins